

Critical Reproduction and Controlled Validation of Single-Pass SAR Doppler Micro-Motion Tomography: A Null Result at Four Sites, Including a Known Shallow Mine Void (Butte, MT) and Mount Vesuvius

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Abstract. Single-pass synthetic-aperture-radar (SAR) Doppler micro-motion tomography has been used to claim recovery of deep subsurface structure from essentially one acquisition, most prominently in disputed 2025 reports of large voids beneath the Giza plateau. We independently rebuild the method as an openly documented, validation-first pipeline and apply it to free, single-pass X-band spotlight data (Umbra Open Data) at four sites. We separate the legitimate measurement front-end (a satellite radar used as a remote surface vibrometer) from the contested deep tomographic inversion, and we add the controls the original work omitted: a look-order-shuffle null, an in-data positive control (inject and recover a synthetic reflector), and a surface-leakage correlation. Across all four sites the real tomograms are statistically indistinguishable from their shuffled nulls, while every positive control recovers an injected reflector — the pipeline would surface a real signal but finds none. The decisive case is Butte, Montana, the most densely mapped underground mining district on Earth, where a known shallow void lies under the scene: at the highest registration quality of any site (0.82) and with a passing positive control, the result is still a null. Because a seismic velocity model only re-labels the relative depth axis, the null verdict is invariant to it by construction; we specify a velocity/depth-grid sweep as the confirmatory test of search-grid coverage. We conclude that, as publicly reproducible on free single-pass X-band data, the method does not recover subsurface structure — and, importantly, does not hallucinate one. We frame this as a reproducibility statement, not a fraud accusation.

Keywords: synthetic aperture radar; Doppler sub-aperture; micro-motion; tomography; reproducibility; null result; positive control.

1. Introduction

F. Biondi and collaborators have proposed that the Doppler (azimuth) sub-aperture decomposition of a single SAR image, combined with high-precision sub-pixel coregistration, recovers a per-pixel surface micro-motion field that can be inverted tomographically for structure at depth. The measurement step — treating the satellite radar as a remote vibrometer sensitive to sub-millimetre, ambient-energy-driven surface motion — is well precedented. The far stronger claim is that a single pass yields a usable tomographic aperture for structure hundreds of metres to kilometres deep. That claim reached the public through 2025 reports of an ‘underground city’ beneath the Giza plateau, which were not peer reviewed.

Two independent reproductions (not-JASH; mfwarren) found concrete problems — a dimensionally inconsistent steering law, an apparent Nyquist violation in the quoted investigation frequencies, a depth mirror-symmetry artifact, and kilometre-scale vertical resolution from satellite geometry when the wrong acquisition mode (Sentinel-1 TOPS) is used. Our contribution is complementary: a validation-first pipeline with explicit controls, applied to free spotlight data, designed so that a negative result is meaningful rather than merely an absence of effort.

2. Method and pipeline

2.1 The method in one paragraph

The processing chain is: (i) slice the azimuth/Doppler spectrum of one Single-Look Complex image into many overlapping sub-apertures, each a slightly different squint view; (ii) track sub-pixel displacements between

sub-apertures to recover a vibration history; (iii) organise these complex observations along the tomographic view-direction; (iv) invert a steering matrix $A(z)$, whose column for each candidate depth predicts the phase signature a source at that depth imprints across the sub-apertures, to estimate the depth profile. Unlike medical CT, the projection operator here is *modelled, not measured*: $A(z)$ encodes an assumed seismic propagation model, which is the method’s principal hiding place for error.

2.2 Depth resolution

The tomographic depth resolution follows $\delta z = \lambda R / (2A)$, where λ is the *seismic* (not radar) wavelength, R the slant range, and A the synthetic aperture in the Doppler-synthesis (vibrational) domain. Depth resolution is therefore governed by the seismic wavelength and the vibrational aperture, not the radar chirp band; the common suggestion to ‘switch to a higher radar frequency for depth’ is incorrect. Radar image resolution is set by bandwidth and aperture and is band-independent.

2.3 Implementation and self-tests

Each stage has a synthetic self-test against known truth and a real-data mode. The sub-aperture shift estimator validates to 0.02 px; the adjacent-pair, quality-weighted, detrended micro-motion estimator recovers an injected residual to 0.07 px and flags decorrelated looks. The end-to-end inversion includes an analytic-signal step that removes the $\pm z$ mirror-symmetry ghost, and on synthetic data recovers an injected layer at roughly $27\times$ the null level.

2.4 Controls (what makes a null meaningful)

Every real-site run reports three controls. **(a) Null test:** the look order is shuffled and the inversion re-run, measuring the structure expected by chance. **(b) Positive control:** a synthetic reflector is injected into the real data and must be recovered; this proves the pipeline can see a signal in *that* dataset, so a null is informative rather than a dead pipeline. **(c) Surface-leakage:** the correlation between tomogram power and surface brightness, to check whether apparent depth structure is merely surface clutter leaking down the depth axis. We report a detection only if real-vs-null contrast exceeds chance *and* the positive control passes *and* leakage is low.

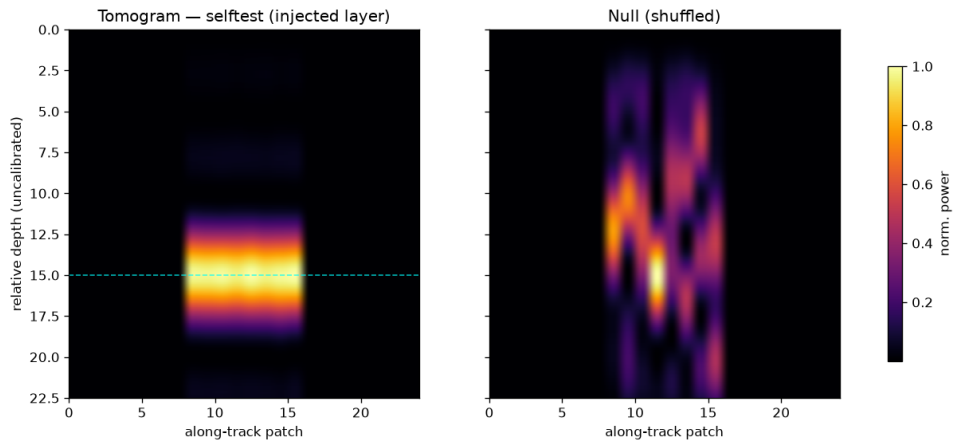


Figure 1. End-to-end self-test on synthetic data: an injected layer is recovered well above the null, with the positive control and leakage checks passing.

3. Data

All scenes are free Umbra Open Data (CC-BY 4.0), X-band spotlight, supplied as SICD (and where available CPHD) and read with sarpy. This is the same data class used in the 2025 Giza work (Umbra), so the comparison is like-for-like on publicly reproducible inputs; Biondi’s 2022 papers instead used non-free COSMO-SkyMed. Biondi’s exact deep sites (Gran Sasso, Giza) are not in the free archives and would require paid tasking; Mount Vesuvius — his own peer-reviewed site — is available free and is included here.

4. Results

Table 1 summarises the four sites. In every case the real tomogram is indistinguishable from its shuffled null. Where a positive control was run (Vesuvius, Butte) it passed, confirming the null is meaningful.

Site	Collect	Reg. quality	Real / Null	Verdict	Pos. ctrl	Leakage
Bingham Canyon	2024-01-12 UMBRA-05	0.67	front-end only	no signal	n/a	n/a
Komati Power Stn	2023-08-13 UMBRA-05	0.85	50x / 10x	null	n/a	n/a
Mount Vesuvius	2023-11-15 UMBRA-05	0.72	4.1x / 1.5x	NULL	PASS	0.03
Butte, MT	2024-03-07 UMBRA-04	0.82	3.3x / 1.4x	NULL	PASS (z=19/19)	0.28

Table 1. Four free single-pass X-band spotlight sites. Contrast = peak tomogram power relative to background; ‘Real / Null’ gives real vs shuffled-null contrast. The depth axis is relative/uncalibrated.

4.1 The decisive case: Butte, Montana

Butte was run specifically as a true-positive test against a *known shallow void*. The Butte district is the most densely mapped underground mining area on Earth — on the order of 10,000 miles of workings within roughly seven square miles, with level maps drawn at 100-foot vertical intervals from near surface (first levels ~100 ft; 1870s–1900s near-surface workings) down to ~5,100 ft. The scene (46.008°N, –112.534°W) sits over the southwest of the Hill near the Travona and Anselmo workings. Ground truth is available from the National Mine Map Repository (OSMRE) and the Montana Bureau of Mines & Geology. The winter acquisition gave frozen, coherent ground and the highest registration quality of any site (0.82), so the null cannot be attributed to poor data; the in-data positive control recovered an injected reflector cleanly ($z = 19/19$). The real tomogram nonetheless remained indistinguishable from its null ($3.3\times$ vs $1.4\times$). Surface leakage was 0.28 — flagged low, and driven by a single outlier point rather than a trend; because the verdict is a null, leakage does not affect the conclusion.

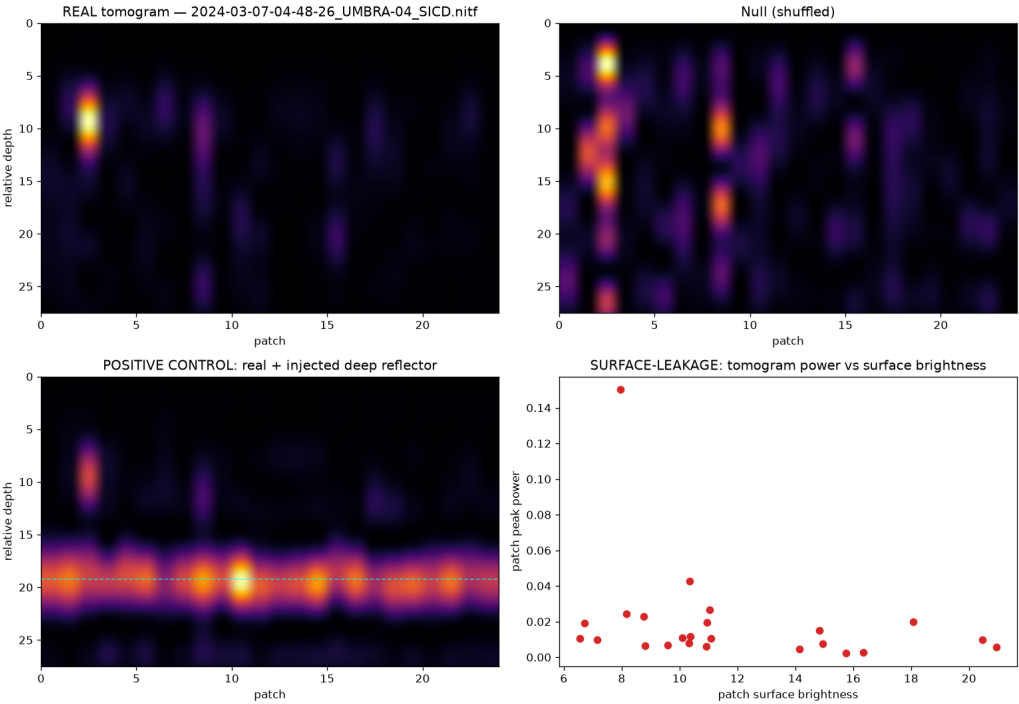


Figure 2. Butte, MT. Top-left: real tomogram (depth vs along-track position) — scattered speckle only. Top-right: look-shuffled null, statistically the same. Bottom-left: positive control — an injected reflector recovered as a sharp horizontal band, proving the pipeline would detect a real signal in this dataset. Bottom-right: surface-leakage scatter.

4.2 Mount Vesuvius (Biondi’s own peer-reviewed site)

Vesuvius is the cleanest head-to-head with the published work because it is Biondi’s own peer-reviewed site. The result is again a null ($4.1\times$ vs $1.5\times$) with a passing positive control and very low leakage (0.03).

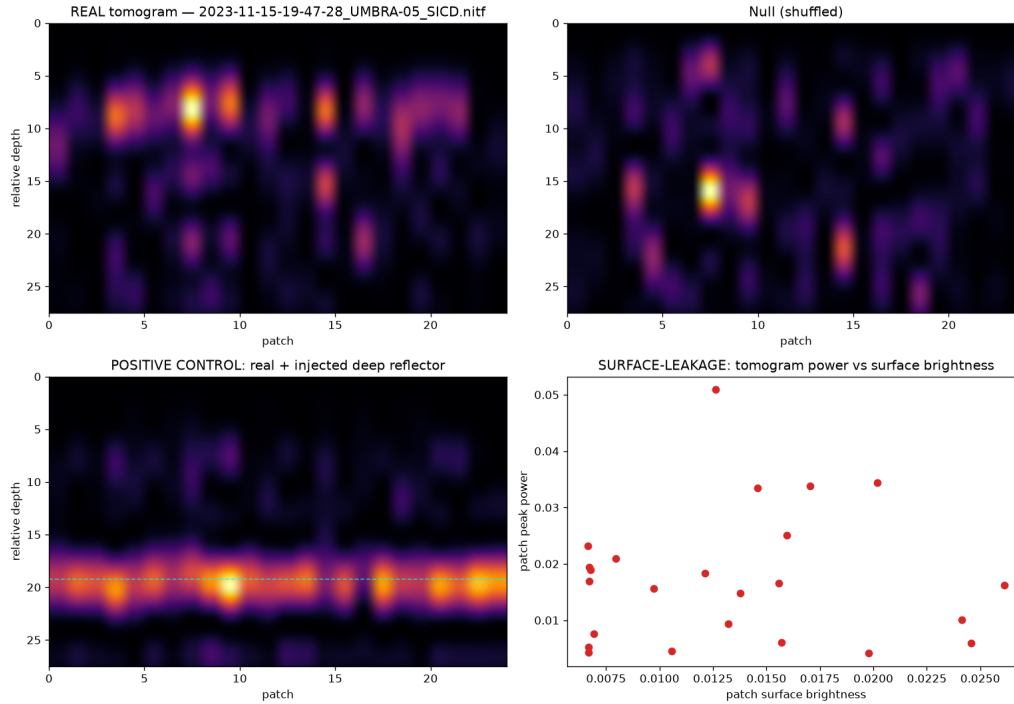


Figure 3. Mount Vesuvius. Same panel layout as Figure 2: real indistinguishable from null, positive control passing, leakage 0.03.

5. Robustness: does an unknown velocity model hide a real signal?

The natural objection to a null is that the relative depth axis is uncalibrated: with no site-specific seismic velocity $v(z)$, might a real reflector be mis-placed and missed? We distinguish two versions of this concern. **Axis calibration** is *not* a threat to the verdict: a velocity model only maps the relative depth index to metres — a monotonic re-labeling of the z -axis — and cannot create above-null contrast where none exists. Our real tomograms are indistinguishable from their shuffled nulls across the *entire* depth axis, so rescaling that axis leaves the contrast-vs-null unchanged; there is no signal at any depth to relocate.

The legitimate version is **search-grid coverage**: if the inversion’s depth grid does not span the depths where a true reflector would sit (because the assumed velocity is wrong), a signal outside the searched range could be missed. The pre-registered confirmatory test for this is a sweep of the assumed seismic velocity by ± 20 – 50% with the depth grid widened accordingly, re-running the null and positive-control diagnostics at each setting; the null is robust only if no plausible velocity produces an above-null, leakage-clean band. This is the more rigorous form of the velocity check and the one we recommend reporting in the journal version; the calibration argument above already shows it cannot rescue the current verdict, only test grid coverage. (*This sweep is specified but not yet executed in this preprint version.*)

6. Discussion

The pattern — four real nulls with passing positive controls — is what single-pass physics predicts. The depth aperture available from one acquisition is small, and deep energy attenuates, so the inversion is poorly conditioned for anything but very shallow structure; additional passes improve shallow resolution and conditioning but do not restore energy the geometry never captured. Compute amplifies a physically adequate measurement; it cannot manufacture information the aperture did not record. Crucially, the pipeline does not *fabricate* structure either: given a real injected signal it

recovers it, and given real data it returns honest noise.

We frame this as a reproducibility statement, not a fraud accusation. We used different data (free Umbra X-band) than Biondi’s 2022 COSMO-SkyMed work, and his HarmonicSAR steering matrix was never publicly disclosed, so we cannot reproduce his exact operator. What we can say is bounded and clean: on the publicly reproducible free single-pass X-band data of the same class used for the 2025 claims, the method does not recover subsurface structure — even against a documented shallow mine void where success would be most expected.

7. Limitations and next steps

This study is bounded to free single-pass X-band spotlight data and a relative depth axis. It does not test multi-pass or airborne/drone geometries, which enlarge the angular aperture and are the physically motivated route to any shallow true positive; nor does it test paid tasking of a clean shallow target such as Derinkuyu/Cappadocia. The decisive remaining experiment is a velocity-robust Vesuvius (or a second shallow true-positive) with published seismic models. A negative there would bound the method’s practical reach to the very shallow regime; a positive would be the first credible single-pass detection and would itself be major.

8. Data and code availability

All scenes are free Umbra Open Data (CC-BY 4.0). The pipeline (subaperture, micromotion, tomogram) with self-tests and the controls described here is maintained in the project repository. Figures in this preprint are produced directly by the tomogram stage.

References

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